CHENNAI METRO RAIL LIMITED (CMRL)

INDUSTRIAL INTERNSHIP PROJECT REPORT

DESIGN AND DEVELOPMENT OF AN AI-POWERED CONTRACTOR EVALUATION SYSTEM AND CHATBOT

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1. EXECUTIVE SUMMARY

During the intensive one-month on-site internship at Chennai Metro Rail Limited (CMRL) from March 16, 2026, to April 16, 2026, a production-ready, localized artificial intelligence system titled 'CMRL Contractor Evaluation System' was conceptualized, developed, and evaluated. Operating under the rigorous guidelines of the Safety Department and under the direct supervision of Thiru Sundrarajan, DGM (Safety), this project solves a foundational bottleneck in industrial contract procurement: the rapid, objective, and multi-parametric screening of engineering and construction contractors. The application relies on advanced natural language processing (NLP) and rule-based decision-making. By leveraging Python, Streamlit, PyPDF, and a localized deployment of the state-of-the-art Gemma 3 large language model through Ollama, the system enables non-technical administrative officers to drop unstructured PDF portfolios into an interactive portal and query specific contractor needs using fluent, conversational human English. The core codebase automatically extracts company details, handles system state across multi-tenant uploads, parses intents with the LLM, filters matching records, and runs a strict mathematical scoring matrix. This report covers the full end-to-end technical documentation, underlying algorithms, code architecture, and deployment paradigms for CMRL's digital infrastructure enhancement.

2. INTRODUCTION TO CHENNAI METRO RAIL LIMITED

2.1 Organizational Background

Chennai Metro Rail Limited (CMRL) is a joint venture established between the Government of India and the Government of Tamil Nadu, responsible for implementing, maintaining, and expanding the rapid transit railway network serving the metropolitan hub of Chennai. Operating under strict ISO 9001:2015 and ISO 14001:2015 certifications, CMRL demands the absolute peak of operational safety, structural integrity, environmental compliance, and cost efficiency across all phases of engineering design and civil execution. The expanding footprint of Phase II lines necessitates massive multi-billion-rupee procurements involving thousands of diverse commercial contractors.

2.2 Role of the Safety Department in Procurement

Within CMRL, the Safety Department holds ultimate regulatory and oversight power. No vendor or infrastructure consortium can be awarded critical trackway, tunneling, electrification, or civil construction contracts without meeting rigid benchmark thresholds regarding safety compliance ratings, past project execution excellence, and technical experience history. The vetting process mandates reading hundreds of pages of technical portfolios, audit records, and corporate disclosures. Ensuring that zero human oversight occurs during this critical filtering step is paramount to avoiding severe operational risks, structural incidents, or project delays.

3. PROBLEM STATEMENT & PROJECT SCOPE

3.1 Existing Contractor Screening Workflow

Historically, the process of evaluating corporate vendor submissions involves human specialists manually reviewing heavy physical binders or digitized unstructured PDF records. For each tender notification, numerous contractors provide profiles detailing their financial proposals, engineering backgrounds, counts of successfully completed massive transport projects, and regulatory safety scorecards. This manual reading cycle severely limits operational agility, consuming dozens of labor-hours per procurement batch.

3.2 Vulnerabilities and Inefficiencies

The legacy human-centric system exhibits several systemic failure modes:
1. Cognitive Fatigue: Reviewers manually scrolling through hundreds of textual lines are susceptible to overlooking key risk factors, such as low safety ratings buried in financial summaries.
2. Subjective Bias: Different evaluators define terms like 'highly experienced' or 'low cost' inconsistently, introducing variance into the vendor selection logic.
3. Scalability Barriers: As Phase II scales rapidly, the volume of proposals grows exponentially, stalling prompt evaluations.

3.3 Scope of the Automated Solution

This project bridges the gap by building an isolated, highly performant, automated AI evaluation system. The software takes raw PDFs, maps natural language intents into technical logical constraints, applies rigorous multi-attribute sorting criteria, and presents an unbiased, auditable recommendation matrix within a clean web workspace. This completely replaces manual document sorting, allowing the Safety Department to complete multi-vendor initial screenings in under ten seconds.

4. SYSTEM ARCHITECTURE AND TECHNICAL STACK

4.1 Architectural Design

The CMRL Contractor Evaluation System is built as a micro-monolithic local architecture optimized for deployment inside an air-gapped network environment. Data privacy is preserved by ensuring that no document content or user query leaves CMRL’s local hardware. The multi-tiered architecture follows a linear, non-blocking asynchronous pipeline:

1. Presentation Layer (Streamlit): Handles multi-file drops, keeps real-time system states, and streams back chatbot conversations.
2. Data Processing Layer (PyPDF Engine): Decodes binary stream files, aggregates textual character grids, and parses structured key-value maps.
3. Semantic Layer (Ollama Integration): Evaluates human intent from input phrases using the localized gemma3 7B LLM engine without cloud dependencies.

4. Analytical Logic Layer (Rule Matrix): Evaluates algorithmic weights, aggregates scores, and runs multi-parameter vector sorting.

4.2 Framework and Component Selection

The technology choices were guided by performance, security, and integration simplicity. Streamlit was selected for its native layout handling and atomic reactive rendering, eliminating complex JavaScript frontend pipelines. PyPDF provides ultra-fast text block generation without heavy external machine learning overhead. Ollama was picked as the local inference model tool due to its low VRAM usage and optimized Gemma 3 quantizations, delivering sub-second model responses for complex text parsing tasks.

5. DETAILED IMPLEMENTATION & MODULE ANALYSIS

5.1 Core PDF Processing and Text Parsing Engine

The data ingestion pipeline utilizes the `pypdf.PdfReader` class. Upon receiving binary byte arrays from the frontend widget, the engine initializes a reader loop across every indexed document page. The text content is compiled sequentially and standard line tokenization is triggered. The parser loops over every line, searching for specific string qualifiers followed by a colon divider.
A clean python structure is initialized for each contractor containing safety, project history, cost, and experience values. The code implements try-except blocks around numerical type conversions. This prevents crashing when contractors input strings or

unreadable characters into numerical fields, falling back to safe default values automatically.

5.2 Application State and Multi-File Session Management

Streamlit re-runs the entire python script upon any user click or text entry. To prevent expensive document re-parsing and infinite loops, the system implements `st.session_state`. The chat history list and parsed data structures are committed directly to session state cache. This guarantees that files are loaded and parsed only once per execution context, keeping chat sub-second and preserving memory allocation bounds.

5.3 Large Language Model Integration via Ollama & LangChain

The application leverages the `ChatOllama` wrapper from the `langchain_community` suite. When a user sends a query, a precise instructions prompt template is populated. It contains few-shot learning examples mapping phrases like 'low cost' to explicit programmatic rules such as `cost < 50`. Gemma 3 acts as an advanced intent interpreter, providing standardized boolean-like rules that the code can parse linearly using basic string searches.

6. MATHEMATICAL SCORING METHODOLOGY

6.1 Evaluation Parameters

To eliminate manual evaluation bias, every validated contractor is evaluated using a rigorous composite multi-attribute calculation. Four core metrics form the vector score: technical tenure experience, tender project cost, industrial safety rating, and successful history of major rail infrastructure completions.

6.2 Mathematical Formula Formulation

The total score S is calculated dynamically using a multi-conditional linear equation. Let E be the years of experience, C be the project cost in standard units, SR be the corporate safety rating (1-5 scale), and P be the number of past successfully completed projects. The programmatic algorithm maps to this formal mathematical criteria:

• If E > 5 years, then Score_Experience = 30, else 0.

• If C < 50 units, then Score_Cost = 30, else 0.

• Score_Safety = SR * 5 (Maximum possible = 25 points).

• Score_Projects = Minimum(P * 2, 20) (Maximum possible capped at 20 points).

Total Composite Score (S) = Score_Experience + Score_Cost + Score_Safety + Score_Projects

Evaluation Metric	Programmatic Boundary Condition	Maximum Weight Allocable
Technical Tenure (Experience)	Experience > 5 Years	30 Points
Financial Tender (Cost)	Cost < 50 Financial Units	30 Points
Operational Safety Score	Safety Rating (1 to 5 Scale) * 5	25 Points
Project Infrastructure History	Min(Projects Completed * 2, 20)	20 Points

7. VERIFICATION, RESULTS & DISCUSSIONS

7.1 Sample Query Test Cases

The application was thoroughly verified using mock data arrays mirroring standard CMRL contract applications. In testing, users input diverse unstructured query scenarios. For example, when a user enters the query 'low cost contractor with high safety', the integrated model engine correctly parses the text inputs without losing context.

7.2 Rule Generation Analysis

The LLM responds by returning standardized rules: `cost < 50 AND safety > 3`. The application filtering code processes these matching substrings and isolating compliant entries. The math engine then applies the composite scoring calculation to rank the contractors transparently. The best vendor is featured prominently at the top of the interface, providing the procurement team with a clear, direct recommendation.

8. CHALLENGES, LIMITATIONS & FUTURE ENHANCEMENTS

A main challenge identified during development was the handling of unformatted PDF text arrays. If a contractor submits a profile as a scanned image rather than a text-based PDF, basic parsing fails to extract characters. To address this, integrating a local optical character recognition (OCR) engine like Tesseract is planned for the next development phase.
Future updates will also expand the application from processing single files to connecting directly with live procurement databases (such as PostgreSQL or Oracle systems). This will allow the system to cross-reference contractor submittals with historical performance records automatically. Additionally, adding interactive data dashboards with charts like radar plots will provide a more detailed, visual view of multi-contractor performance profiles.

9. CONCLUSION

The CMRL Contractor Evaluation System successfully demonstrates how local artificial intelligence models can streamline complex administrative workflows. By combining modern language processing with a stable, objective mathematical scoring matrix, the tool automates vendor vetting while fully protecting sensitive corporate data. Developed under the supervision of the Safety Department, this system provides CMRL with an effective, scalable proof-of-concept for modernizing infrastructure procurement workflows.

10. REFERENCES & APPENDIX

1. Streamlit Documentation: Core Session State and Ingestion APIs, 2025.
2. LangChain Community: Local LLM Orchestration and Prompt Templates, 2026.
3. Ollama Engine: Running Quantized Language Models on Edge Devices, 2026.
4. Chennai Metro Rail Limited (CMRL): Internal Safety Procurement Standards and Technical Directives, 2026.